LightBanglaHC: Evaluation Validity, Inference Reconstruction, and Accounting Boundaries in Harmful-Content Classification

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Abstract. Resource-conscious evaluation of harmful-content classifiers must distinguish a small classifier component from the complete predictor: the efficiency of a sparse system cannot be established by counting a combination head while excluding the fitted representations and estimators that produce its inputs. We examine this alignment problem through a retrospective case study of LightBanglaHC, a recorded implementation associated with the HerWILL AI for Digital Safety Datathon 2026, whose notebook records 47,817 training examples, 11,955 test rows, three implementation-defined classes, and 254,983 concatenated TF–IDF features feeding four classical classifier families and two alternative linear stackers. A PyTorch linear head attains the highest selected pooled out-of-fold macro-F1, 0.5295, against 0.5246 for logistic regression, but the difference is descriptive rather than independently validated: globally fitted transformations and class weights, nonnested meta-validation, and checkpoint and candidate selection connect fitting to the reported outcomes, and a source-level dependency analysis explains why row-wise out-of-fold exclusion does not isolate the complete stacking procedure. Parameter accounting establishes 39 scalars in one head and 195 across the five heads averaged at test time, while the five logistic-regression fits on the same path imply 3,824,760 coefficients and intercepts. Separating source observations, recorded outputs, arithmetic reconstructions, unverified external scores, and prospective experiments, we specify a corrected evaluation and measurement protocol. The contribution is an inspectable case study of inference, evaluation, and accounting alignment, not a new algorithm or a demonstrated deployment advantage.

Keywords: Harmful-content classification · Evaluation validity · Stacked generalization · Accounting boundaries · Reproducibility.

1 Introduction

Harmful-content classification joins a predictive problem to a problem of interpretation: labels encode decisions about which expressions count as harmful and when. Offensive vocabulary, hostility toward an individual, and attacks on a social group are not interchangeable constructs, as the distinction between hate speech and offensive language [6] and the separate dimensions of directness, hostility type, and target [14] show. Because Bangla text classification already spans sparse models, ensembles, pretraining, and code-mixed representations [19, 22, 3, 17], a sparse system cannot be motivated by an absence

of alternatives; what is needed is precision about the representation, prediction path, and resource boundary under comparison.

We frame this as alignment among three objects: the *inference object*, the complete computation turning raw text into the reported prediction; the *evaluation object*, the training-and-selection procedure whose predictions the reported metric evaluates; and the *accounting object*, the delimited set of artifacts included in a cost statement. Counting a neural head omits the vocabulary, IDF weights, calibrated base estimators, and additional heads its predictions depend on, even though Green AI motivates reporting expenditure alongside accuracy [21]. This matters most in stacking: held-out base predictions are useful training inputs for a second-level learner [25], but their out-of-fold (OOF) status does not make every subsequent use an independent evaluation, because a meta-training feature can depend on labels from a partition later treated as meta-validation. Global fitting and repeated selection create further paths that a fold label does not reveal, which is why work on model-selection bias [24, 5] and leakage [10] motivates inspecting the complete procedure.

The case is LightBanglaHC, developed for the 2026 HerWILL AI for Digital Safety Datathon. We analyse one supplied notebook execution, its source, and preserved aggregate outputs; training is not rerun and the outputs are not authenticated against original rows, since the raw CSVs, row-level predictions, and fitted artifacts are unavailable. Bangla-aware preprocessing establishes neither a Bangla-only corpus nor any language-specific performance, and the class names are code-assigned descriptors, not an annotation manual. The paper contributes an evidence-aware reconstruction of the executed pipeline, a dependency analysis of the selected OOF result, a scoped accounting analysis separating one head from the retained ensemble and upstream state, and a prospective protocol, while claiming neither algorithmic priority, transformer superiority, nor validated resource savings. Claims are tagged **S** (source observation), **R** (recorded output, verifying what the artifact records rather than its authenticity against raw data), **A** (arithmetic reconstruction), **X** (external unverified quantity), or **P** (prospective proposal); missing empirical evidence is marked **[REQUIRES NEW EXPERIMENT]**, missing source, governance, or author information **[AUTHOR TO VERIFY]**, and unsupported conclusions **[NOT ESTABLISHED BY AVAILABLE EVIDENCE]**. These markers are retained deliberately and must not be removed by cosmetic editing.

2 Related Work and the Scoped Research Gap

Taxonomy determines what a classification error means, and neither the three-way English scheme of [6] nor the multi-aspect scheme of [14] maps onto LightBanglaHC's explicit, subtle, and neutral classes: "subtle" is not evidence of a validated sarcasm category. BD-SHS is the closest linguistic context, pairing Bangla hate-speech identification with target and type annotations and evaluating linear SVMs over word and character TF–IDF features [19], but no verified provenance connects it to the HerWILL data, so collection context remains an evidence requirement, not an inherited fact [1]. Sparse classifiers are meaningful controls when their features and estimator variants are specified, since Complement Naive Bayes derives from a distinct complement-class estimation [18] and voting designs use comparable representations for Bengali hate

speech [22]; yet sparsity guarantees neither accuracy nor low deployment cost, because sparse inputs coexist with large coefficient arrays and many retained estimators. Stacked generalization learns a fitted second-level predictor from base-model outputs [25], and one precedent pairs classical first-level learners with a neural second-level classifier for Bangla religious-hate comments [8], establishing lineage rather than equivalence.

Evaluation must also include adaptive choices, since cross-validation used both to choose settings and to estimate their error is biased [24, 5]; probability quality is a separate requirement [13, 23]; uncertainty cannot be recovered by treating folds as independent replications [2, 7]; and reproducibility requires code, reporting, and independent checking together [16], with model cards [12] recording explicit unknowns, as a recent end-user audit of ADMET leaderboard models also illustrates [11]. The gap addressed here is not universal neglect of these topics, but that the interaction among inference scope, evaluation ancestry, and accounting boundaries has not been made inspectable within one recorded harmful-content artifact.

3 Study Design, Evidence, and Dataset Scope

This is a retrospective, single-artifact audit whose unit of analysis is the relationship between an implementation, its recorded outputs, and the claims those materials support. The supplied notebook contains one code cell, 1,248 source lines, an execution count of one, two saved standard-output streams, and an embedded figure, with metadata dating the run to 18 April 2026. We identify invoked fitting, prediction, selection, and saving operations by following their inputs rather than descriptive comments, check saved outputs against identities such as class-support totals and confusion-matrix margins, and assign each claim an evidence type. Findings are restricted to the inspected artifact, whose SHA-256 digests are `5a7ea479…38aeedc0` and `675e2638…6acf1a8287`; its raw CSVs, row-level predictions, and fitted artifacts are absent, so corpus statistics, vocabularies, individual predictions, and reload equivalence cannot be checked.

The task assigns one of three labels to an input text, the saved shapes being (47,817, 3) and (11,955, 2) before the processed-text column is added. The recorded supports (R) are 23,756 Explicit Toxic, 18,163 Subtly Toxic, and 5,898 Neutral, or 49.681%, 37.984%, and 12.335% (A), giving balanced weights 0.670946, 0.877553, and 2.702442 (A) under $w_c = n/(3n_c)$, computed once from all training labels before cross-validation; the 11,955 unlabeled test rows are not an evaluated sample. The competition-named input path identifies an access context, not a population: collection dates, annotation instructions, annotator agreement, duplicate structure, language composition, and licensing and consent records are unavailable, so recovering them is prospective (P) and their absence is an evidence gap rather than a negative finding. The reported metric is the unweighted mean of the three classwise F1 values, pooled after assembling one prediction per training row; because F1 is nonlinear in confusion counts, that value need not equal the mean of fold values.

4 Recorded LightBanglaHC Pipeline

Let x be the original string after missing-value handling. The processed string is $s(x) = C(x) \parallel \texttt{FEATSEP} \parallel H(x)$, where C cleans and H emits synthetic tokens. Crucially H receives x, not $C(x)$, so the cleaner's normalization, lowercasing, and token replacements neither erase the original capitalization and punctuation before feature extraction nor normalize the input used by the lexical rules. The injector holds 42 explicit lexical entries, 14 subtle-pattern expressions, and nine sarcasm-pattern expressions (S), each counted once per matching entry, and emits capped markers for lexical hits, script mixture, capitalization, length, punctuation, emoji, and whole-word religious, gender, or threat cues, with no context or intent validated. Those names are code identifiers, not measured constructs: "this is a salad" triggers the entry `sala`.

For branch r, with $t_{ij}^{(r)}$ the count of retained feature j in processed document i, $\mathrm{df}_j^{(r)}$ its document frequency, and N the number of fitting documents, the documented-default interpretation of the omitted normalization and IDF options gives

$$u_{ij}^{(r)} = \left(1 + \ln t_{ij}^{(r)}\right)\left[1 + \ln \tfrac{N+1}{\mathrm{df}_j^{(r)}+1}\right] \text{ for } t_{ij}^{(r)} > 0, \qquad \phi_r(x_i) = u_i^{(r)} / \|u_i^{(r)}\|_2, \tag{1}$$

and $u_{ij}^{(r)} = 0$ otherwise. The source sets sublinear TF for all branches and smoothed IDF for the word branch but leaves normalization and the character-branch DF and IDF options unset, so historical behaviour and the fitted arrays remain **[AUTHOR TO VERIFY]**.[1] The branches are configured (S) as a word analyzer over non-whitespace tokens with 1–4-grams, a 150,000-feature cap and document frequency in $[1, 0.98]$; a `char_wb` analyzer with 2–6-grams and a 100,000-feature cap; and a `char` analyzer with 1–2-grams and a 50,000-feature cap, both requiring two documents. Their sparse concatenation has the only recorded width $D = 254{,}983$ (R); the caps sum to 300,000, but individual fitted widths are unlogged, so assigning widths 150,000, 100,000, and 4,983 would be unsupported. All three vectorizers are fitted to the entire training corpus before folds are constructed, so ϕ is a global map and later label exclusion is not representation isolation.

Table 1 lists the invoked configurations. Complement Naive Bayes (CNB) uses only `char_wb` features [18]; a source comment ties that restriction to nonnegativity, but all three branches are nonnegative, so the choice is an implementation fact rather than an ablated design. Isotonic calibration of CNB, LinearSVC (SVC), and SGDClassifier (SGD) changes the probability-generating procedure supplied to the meta-model [13] and inherits globally fitted features and class weights, and no diagnostic is recorded [23].

Let $V_1, \ldots, V_5$ be disjoint validation sets partitioning $I = \{1, \ldots, n\}$ from shuffled stratified five-fold splitting with seed 42, and let $f(i)$ be the fold containing i. For family b, $h_b^{(-f)}$ is its probability-producing fit on $I \setminus V_f$, including calibration where applicable, with the global ϕ as input; the superscript excludes V_f from estimator fitting, not from vocabulary estimation or weight computation. The meta-training row and the test-time

[1] Consulted API: https://scikit-learn.org/1.5/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html; this documentation version is not evidence of the historical runtime version.

Table 1. Verified invoked configurations (S). Unspecified arguments retain library defaults; installed library versions and fitted calibrator internals are unavailable.

Candidate	Inputs	Explicit settings and training controls
LR	Full TF–IDF	$C = 8$; `saga`; multinomial; max. 2,000 iterations; tolerance 10^{-4}; global balanced weights; seed 42.
CNB	`char_wb`	$\alpha = 0.05$; `norm=True`; isotonic calibration, CV = 3; no explicit class-weight argument.
SVC	Full TF–IDF	$C = 0.3$; max. 5,000 iterations; `dual=True`; global balanced weights; seed 42; isotonic calibration, CV = 3.
SGD	Full TF–IDF	Modified-Huber loss; $\alpha = 10^{-4}$; max. 200 iterations; tolerance 10^{-4}; global balanced weights; seed 42; isotonic calibration, CV = 3.
Meta-LR	12 probabilities	$C = 10$; max. 500 iterations; multinomial; global balanced weights; seed 42; solver not explicitly overridden.
TinyLinear	12 probabilities	Linear(12,3); Xavier-uniform weights; weighted cross-entropy; AdamW rate 0.005, decay 10^{-4}; cosine scheduler $T_{\max} = 30$; max. 30 epochs; patience 8; batches 512/1,024; seeds 42–46.

feature vector are

$$z_i = \left[h_b^{(-f(i))}(\phi(x_i))\right]_{b=1}^{4} \in \mathbb{R}^{12}, \qquad \bar{z}(x) = \left[\tfrac{1}{5}\sum_{f=1}^{5} h_b^{(-f)}(\phi(x))\right]_{b=1}^{4}, \tag{2}$$

so training rows receive one fold-specific base prediction per family while test rows receive within-family averages. TinyLinear computes logits $\ell_i = Wz_i + b$ with $W \in \mathbb{R}^{3\times 12}$ and no hidden layer, trained with weighted cross-entropy; that objective is not macro-F1, which is used instead for checkpoint selection, with every epoch assessed on the same fold later included in reported validation performance. Final test labels are $\widehat{y}(x) = \arg\max_c \frac{1}{5}\sum_{f=1}^{5} q_f(\bar{z}(x))_c$ over the selected heads q_f: five softmax outputs averaged, not one head obtained by averaging coefficients, implying 39 scalar weights and biases per head and 195 across five heads (A) before upstream state. Figure 1 summarises that computation, which is not fully nested: when V_f is a meta-validation set, a meta-training row in another V_g was represented by a base fit using labels from V_f.

5 Experimental Record

The record contains six candidate evaluations under one seeded partitioning scheme; neural seeds 42–46 vary initialization across folds and are not independent repeated evaluations, and Table 2 preserves pooled and fold metrics at their printed precision. TinyLinear is the selected candidate and LR the strongest base learner, with a rounded difference of 0.0049 macro-F1 (A) that is not evidence of stable superiority and does not isolate whether it arose from probability combination, class weighting, regularization, or stopping. No hyperparameter search beyond fixed configurations, epoch selection, and the six-way comparison is recorded, and unused definitions are not experiments: the untrained 12–8–3 MLP has 131 parameters by arithmetic, not the 123 of its comment, and the unused helper rate 0.01 must not replace the invoked 0.005.

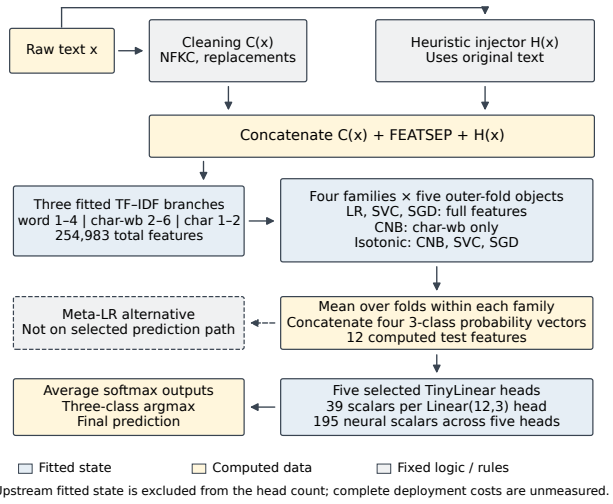


Fig. 1. Source-derived schematic (S), not an experimental plot; the selected path averages five TinyLinear softmax outputs over within-family averages of base-fold predictions.

Table 2. Recorded pooled OOF macro-F1 and per-fold values (R). All candidates share global feature fitting; stackers reuse nonnested OOF features. Folds are dependent partitions, so no interval is attached.

Candidate	Role	Pooled	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
LR	Base learner	0.5246	0.5265	0.5171	0.5259	0.5254	0.5278
CNB	Base learner	0.4478	0.4501	0.4397	0.4573	0.4468	0.4450
SVC	Base learner	0.5053	0.5042	0.4950	0.5152	0.5094	0.5027
SGD	Base learner	0.4977	0.4965	0.4922	0.5049	0.5025	0.4919
Meta-LR	Alt. stacker	0.5268	0.5235	0.5183	0.5220	0.5332	0.5369
TinyLinear	Selected stacker	0.5295	0.5264	0.5255	0.5266	0.5338	0.5350

Table 3 reproduces the nine integer annotations of the recorded confusion figure with classwise metrics reconstructed from them; row sums match all printed supports and total 47,817, and the macro-F1 definition yields pooled 0.529547 with accuracy 0.563544 (A). Neutral has the lowest classwise F1 although Subtle has lower recall, the largest off-diagonal counts are Explicit-to-Subtle (6,406) and Subtle-to-Explicit (6,217), and of 8,841 predicted-Neutral examples 5,595 carry a toxic label while 2,652 actual-Neutral examples are assigned one; these aggregates establish no cause. An always-Explicit predictor would give MacroF1 = $2n_0/[3(n+n_0)] = 0.221275$ (A), an arithmetic reference rather than a measured baseline, and test predictions number 5,501 Explicit, 4,139 Subtle, and 2,315 Neutral (R), which without test labels is neither accuracy nor evidence of a recovered class balance. The source writes a submission with a constant `parameters` field of 39 without loading the configured sample-submission file, so its save message demonstrates neither schema acceptance nor rank; upstream transforms

Table 3. Selected TinyLinear OOF confusion matrix (R) with diagonal row percentages and class-wise metrics computed from those counts (A). Rows are actual, columns predicted labels; class-wise values are rounded from the exact count-derived figures.

Actual	Explicit	Subtle	Neutral	Supp.	Prec.	Rec.	F1
Explicit	15,192 (63.95%)	6,406	2,158	23,756	0.6790	0.6395	0.6587
Subtle	6,217	8,509 (46.85%)	3,437	18,163	0.5125	0.4685	0.4895
Neutral	964	1,688	3,246 (55.04%)	5,898	0.3672	0.5504	0.4405
Predicted (A)	22,373	16,603	8,841	47,817	Macro-F1	0.529547 (A)	

are not serialized, the device string and Python metadata conflict, and library versions, selected epochs, and fitted-state bytes are missing.

6 Evaluation Validity

The recorded macro-F1 answers a descriptive question—which predictions this notebook retained and scored—rather than how a fully specified training-and-selection procedure performs on untouched examples. Three requirements differ: exclusion of a row from its own base fit, exclusion of evaluation information from the whole fitting and selection ancestry, and assumptions sufficient for uncertainty estimation. Three dependency paths are source-supported. First, all vectorizers are fitted before the fold loop, so the validation covariates X_{V_a} contribute to vocabulary membership, document-frequency eligibility, cap-based selection, and IDF values, which fixing hyperparameters does not fix; because test text is only transformed, this is validation-covariate exposure within the labeled corpus. Second, the global class weights $w_{I,c} = |I|/(3n_{I,c})$ depend on class counts in every fold. Third, and most consequentially, for a meta-learner $q^{(-a)}$ nominally trained on $\{(z_i, y_i) : i \notin V_a\}$ and a meta-training row $i \in V_b$ with $b \neq a$, the base fitting set $I \setminus V_b$ includes V_a, so

$$y_{V_a} \longrightarrow h_m^{(-b)} \longrightarrow z_i \longrightarrow \text{meta-training data} \longrightarrow q^{(-a)}. \tag{3}$$

A row $j \in V_a$ does receive its own base features from fits excluding V_a, but that exclusion is insufficient: the meta-model evaluating j was trained on features whose construction used y_{V_a}, a pathway that persists unless the stacking features are rebuilt inside the outer-training partition. The graph establishes a source-level violation of full-procedure holdout, not the size or sign of its effect.

Adaptive stopping and candidate selection add exposure: TinyLinear's checkpoint is selected using the labels later used to score it, and the six-way comparison selects a winner using pooled OOF outcomes without creating a new evaluation sample—opportunities for selection optimism [5, 24] rather than a measured bias correction. Keeping outer labels untouched is also distinct from making fold scores independent, since training sets overlap and stratification controls class balance rather than shared-user, duplicate, or temporal dependence; treating five scores as i.i.d. observations is therefore inappropriate, and this audit reports neither a significance test nor a confidence interval.

7 End-to-End Parameter and Resource Accounting

The measurement boundary must match the computation whose predictions are evaluated. The selected test procedure transforms raw text, obtains probabilities from LR, CNB, SVC, and SGD, averages five outer-fold outputs within each family, applies five neural heads, and takes an argmax of their averaged softmax outputs; Meta-LR is an alternative candidate rather than a component, so counting unused candidates would overstate the requirement while omitting upstream predictors would understate it. Assuming the stated 3×12 coefficient matrix with intercept per head and $D = 254{,}983$,

$$\begin{aligned} N_{\text{head}} &= 3 \times 12 + 3 = 39, & N_{5\text{ heads}} &= 5 \times 39 = 195, \\ N_{\text{LR}} &= 3D + 3 = 764{,}952, & N_{5\text{ LR}} &= 5(3D + 3) = 3{,}824{,}760. \end{aligned} \tag{4}$$

These are scalar counts for specified structures, not interchangeable definitions of model size; averaging five softmax outputs is not equivalent to the softmax of averaged coefficients, so a single-head deployment would be a different evaluated object unless prediction equivalence were established.

Beyond these counted objects the inventory is open rather than zero: the fitted vectorizers retain vocabulary mappings and IDF arrays, and no serialized artifact, RAM figure, or benchmark exists for the complete predictor. Because the notebook serializes no fitted LR objects, the five-fit subtotal is a replay inventory rather than observed memory occupancy—a structural lower bound that cannot be converted into bytes without dtypes, encodings, and the artifacts. CNB, SVC, and SGD each contribute five outer calibrated objects whose internal three-fold calibration can retain multiple fitted classifiers and calibration functions; with library versions and fitted objects unavailable, no complete-system total is asserted, and the unknown realized `char_wb` width prevents an exact CNB inventory [15]. A defensible comparison must report the same boundary for every candidate, since parameters, fitted state, serialized bytes, peak RAM, latency, and energy are distinct quantities [9]. The saved output reads `Manual NOP: 39`, the manual tensor-element fallback rather than a successful profile, and even a successful head profile would omit vectorization, scikit-learn prediction, calibration, and outer averaging. Artifact closure remains open: neither fitted vectorizers nor base estimators are persisted and the checkpoint bytes are absent, so every prediction-time component must be recovered and raw-text equivalence verified after fresh-process loading [16]. The accounting result is thus a mismatch of scopes, not a measured deployment advantage.

Score provenance

Five score categories must also be kept apart. With $F = 0.5295466327432113$, $N = 39$, and $\epsilon = 5 \times 10^{-16}$, the notebook implements $S_{\text{local}} = F/(N + \epsilon) = 0.0135781187883$; an earlier manuscript equation $0.4F + 0.6/[\ln(N + 1) + \epsilon]$ gives 0.3744696715064; and an earlier diagram $F/\log_{10}(N+1)$ gives 0.3305410756212. These are three calculations on one reconstructed F, not measured outcomes; the user-reported leaderboard values, public 0.37286 and private 0.37500, are external reports (X); and the official scoring contract is unknown. Because the local table assigns $N = 39$ to all six candidates, including classifiers whose predictions never traverse that head, ranking by the local score

is exactly ranking by F, and since $0 \le F \le 1$ implies $S_{\text{local}} \le 1/39 \approx 0.0256$, neither leaderboard value can be that ratio; inverting the weighted expression instead yields $F_{\text{public}} \approx 0.5255$ and $F_{\text{private}} \approx 0.5309$, whose proximity to the recorded value verifies nothing, so comparing 0.5295 with 0.37286 does not establish a generalization drop.

8 Discussion, Limitations, and Corrected Prospective Evaluation

The recorded ordering describes a development record rather than a successful stacking experiment, since the difference is measured on predictions selected through different training mechanisms within a shared, non-isolated validation construction; a predictive advantage of the selected procedure is **[NOT ESTABLISHED BY AVAILABLE EVIDENCE]**, and since Meta-LR and TinyLinear both compute affine class scores on the same inputs, a matched comparison should evaluate procedures rather than credit a PyTorch module. The transferable contribution is the discipline of attaching an object, a fitting ancestry, an evidence type, and a boundary to each claim: the counted head is not the raw-text predictor, a base prediction can be OOF while the meta-training table still has a path from evaluation labels, and a plausible formula can reproduce the scale of an external score without identifying its meaning. The three-way argmax is likewise a research label, not a moderation policy, since substring matching can fire inside a benign word and identity-named cues do not model whether a speaker endorses, condemns, quotes, or reclaims an expression [4, 20]: no deployment policy is supported here, **[NOT ESTABLISHED BY AVAILABLE EVIDENCE]**; a contextual error study remains **[REQUIRES NEW EXPERIMENT]**; and no legal basis, consent, or ethics approval is asserted: **[AUTHOR TO VERIFY]**.

The limitations are of two kinds. The original procedure offers no verified untouched final evaluation, matched transformer experiment, fixed-averaging comparison, ablation, calibration measurement, or resource benchmark, so the corrected-procedure experiment remains **[REQUIRES NEW EXPERIMENT]**. The reconstruction has different limits: source and saved output cannot be authenticated against raw inputs, so agreement between transcribed counts and printed supports is internal consistency rather than data authenticity, and unknown package versions limit reconstruction of version-dependent defaults, leaving complete replay **[NOT ESTABLISHED BY AVAILABLE EVIDENCE]**. The threats are distinct in kind—*internal*, because fitting and selection use evaluation information; *external*, because one artifact supports no general claim; *construct*, because label names lack an annotation manual; *statistical*, because folds are dependent and paired predictions absent; and *measurement*, because one countable component differs from the full prediction path.

A corrective study is specified rather than performed: every item below is **[REQUIRES NEW EXPERIMENT]**, and data access, annotation definitions, and release permissions are **[AUTHOR TO VERIFY]**. Before splitting, the authorized dataset must be recovered and hashed, the label mapping identified, near duplicates inspected, and any structure defining dependent units determined; and because the architecture has already been informed by this corpus, the new procedure should be frozen before a rerun and that exposure disclosed. Let E_a be one outer evaluation partition and $T_a = I \setminus E_a$ the permitted training set: every data-dependent operation producing predictions on E_a must

be fitted using T_a only, and moving the three TF–IDF fits into T_a is insufficient if inner validation reuses features or labels from its own assessment set, so the whole pipeline, including its base OOF features, must be rebuilt within each inner candidate's fitting subset, with checkpoint epochs selected by complete-pipeline validation inside T_a. The minimum predictive study is then (A) the fully nested procedure with training-only fitting and selection and (B) standalone LR against fixed probability averaging, Meta-LR, and TinyLinear on identical partitions with declared budgets; component and calibration claims require (C) marker-family removals, (D) single-branch against combined representations, and (E) Brier score, log loss, and binned reliability fitted training-only; measured resource claims require (F) reloading the complete inference graph in a fresh process and reporting prediction equivalence, artifact bytes, peak RAM, and latency; and comparative or population claims require (G) a justified pretrained checkpoint on the same partitions and (H) frozen-system evaluation with verified annotations. Row-level outputs must preserve identifiers, class ordering, probabilities, and fold and model indices, since aggregate matrices cannot recover the joint error pattern a paired comparison needs.

9 Conclusion

This case study reconstructs a recorded harmful-content classifier built from three sparse TF–IDF branches, four probability-producing classical families, and alternative linear stackers. Its 47,817 training examples, 254,983 feature dimensions, and selected pooled OOF macro-F1 of 0.5295 belong to one stored development record and do not establish a generalization advantage over the recorded LR value of 0.5246. One head contains 39 scalar parameters whereas the recorded test procedure averages five heads over substantial upstream state; row-level OOF exclusion cannot establish complete-procedure isolation when evaluation labels remain ancestors of meta-training features; and an external score's proximity to one formula does not verify its semantics. Every predictive or efficiency claim should therefore name the inference object, its fitting and selection ancestry, its measurement boundary, and the evidence connecting them; until an independent evaluation tests that ordering, this remains an inspectable methodological case, not a validated deployment system.

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